

OPTION 1: BASIC DATA HANDLING WITH LLMS

Objective: Explore how a chat-based LLM can assist you in exploratory data analysis. The first goal of this exercise is to become familiar with LLM applicability for tasks involving the generation of quick insights into data analysis. The second goal is to create a potentially re-usable template for skills / prompts that can support exploratory data analysis with small datasets. You can use a simple chat interface for this exercise, or create a customizable LLM functionality (like Claude skill, Custom GPT). To complete the exercise, follow these steps:

Step 1: Select a dataset. Use either an own dataset (no confidential data!) or a publicly available data set, like this from Kaggle: <https://www.kaggle.com/datasets/parve05/customer-review-dataset> . You can also generate a dataset with an LLM.

Step 2: Set up the assistant. Define the objective (“Report descriptives” / “prepare bar chart visualizations”), behavior (e.g., structured, cautious, and transparent reasoning, no fabricated results, ...), and expected results.

Step 3: Explore the data. Your prompt can be used, e.g., to achieve the following (or multiple of these):

- Generate a data analysis plan
- Identify (and evtl. visualize) descriptives
- Compare group performances or suggest group comparisons
- Explore at least one deeper angle (e.g., simple regressions, text analysis, correlations)

Step 4: Document and reflect.

- Document your prompts and results in a one-pager / slide
- Shortly outline: What do you think are the problems with this approach? Which hurdles did you run into?
- Tip: Depending on how satisfied you are, you can document your prompt / settings in the LLM tool or start your own „prompt template library.“

Send a short self-reflection and your results via mail. Blind any information you do not wish to disclose and note shortly if I may use your results as examples on slides for this course.

OPTION 2: GENERATE A PROMPT TEMPLATE FOR A RESEARCH ASSISTANT

Objective: The objective of this exercise is to generate a re-usable prompt draft to quickly set up a personal Research Assistant. This assistant should be able to assist you for one typical research task you have to fulfil regularly, such as improving paragraphs from your current manuscript, replying to revision notes, etc. The goal is to generate a prompt that you can store and then copy + paste to quickly to different LLMs. Follow these steps to complete the exercise:

Step 1: Define Role:

- “You are my Research Assistant in [FIELD].” (specify expertise (methods, theory, target journals).
- “Please behave as follows: [Add behavioral rules] (e.g., No invented citations, Say “I don’t know” if uncertain, Ask clarification questions, Justify key recommendations...)

Step 2: Provide Context

- My research topic: ...
- My typical methods: ...
- My current project: ...
- The desired output format: ... (e.g., “Use only bullets; be concise, critical...).

Step 3: Apply the Assistant.

For a self-selected task, use the Assistant. Examples can be to proofread and improve a self-written paragraph.

Step 4: Compare and reflect

- Apply and adapt the prompt for the task. Compare to an output without this prompt template. How do you assess the quality and usability of the LLM output?
- Tip: If useful, document and store the prompt. You can start with a documentation in your own „prompt template library.“

Send a short self-reflection and your final prompt template via mail. Blind any information you do not wish to disclose and note shortly if I may use your results as examples on slides for this course.